Module-4
DESIGN OF EXPERIMENTS AND OPTIMIZATION METHODS

Design of experiments; Basic experimental designs; Completely randomized design; Randomized block design; Latin square design; Full factorial design - 2^2, 2^3 and 2^k; Fractional factorial designs; Taguchi methodology – Orthogonal array; S/N ratio – Taguchi method for optimization; Response surface method – multi response optimization. Global optimization techniques – evolutionary and swarm algorithms – examples and case studies.

Design of Experiments (DOE)

- Design of Experiments (DOE) is a systematic method used to plan, conduct, analyze, and interpret controlled tests to evaluate the factors that influence a particular outcome or response.
- To determine cause-and-effect relationships between input variables (factors) and output responses efficiently and accurately.

Objectives of DOE

- Identify which factors significantly affect a response.
- Quantify how factors interact.
- Optimize process conditions for desired outcomes.
- Improve quality and performance with fewer trials.

Key Terms in DOE

Term	Meaning
Factor	An input variable that is changed or controlled in an experiment (e.g., temperature, pressure , time)
Level	The specific value or setting of a factor (e.g., 100°C, 150°C).
Response	The measurable outcome affected by factors (e.g., yield, strength).
Treatment	A combination of factor levels used in one experimental run.
Experimental unit	The smallest division of material or subject to which a treatment is applied.
Replication	Repeating an experiment to estimate experimental error.
Randomization	Randomly assigning treatments to avoid bias.
Blocking	Grouping similar experimental units to reduce variability from known nuisance factors

Basic Principles of Experimental Design

Principle	Description
Replication	Increases precision and allows estimation of experimental error.
Randomization	Protects against systematic bias and confounding.
Blocking	Reduces variation by grouping similar experimental units together
Control	Keeping non-experimental factors constant to isolate the effect of actors of interest

Steps in Designing an Experiment

- Define the objective: What do you want to find out?
- Select response variables: What will you measure?
- Select factors and levels: What inputs will you vary and at what settings?
- Choose experimental design: e.g., Completely Randomized Design, Randomized Block Design, Factorial Design.
- Run the experiment: Apply randomization and replication.
- Analyze data: Use ANOVA, regression, or other statistical tools.
- Draw conclusions and validate results.

Common Types of Experimental Designs

Design Type	Description	Example
Completely Randomized Design (CRD)	Treatments are assigned completely at random.	Fertilizer effect on crop yield.
Randomized Block Design (RBD)	Units are divided into blocks to account for variability.	Testing drug efficacy across patient age groups.
Latin Square Design (LSD)	Controls for two sources of variation (row and column).	Machine performance across operators and shifts.
Factorial Design (2^2, 2^3, 2^k)	Tests all possible combinations of factor levels.	Studying effects of temperature and pressure on reaction rate.
Fractional Factorial Design	Uses a subset of full factorial combinations to reduce trials.	Screening major factors in complex systems.

Basic Experimental Designs

- An experimental design is a structured plan for conducting an experiment to study how changes in one or more factors (independent variables) influence a response (dependent variable).
- It helps researchers:
  - Organize the experiment properly
  - Control sources of variation
  - Obtain valid and reliable results with minimal effort and cost

Objectives of Experimental Design

- To determine the effect of different treatments or factors on a response variable.
- To reduce experimental error and increase the accuracy of results.
- To allow valid statistical analysis (like ANOVA).
- To ensure results can be generalized beyond the experimental conditions.

Principles of a Good Experimental Design

- Replication – Repeating treatments to estimate experimental error.
- Randomization – Assigning treatments randomly to eliminate bias.
- Local Control (Blocking) – Grouping similar experimental units to minimize variation due to nuisance factors.

Completely randomized design

- The Completely Randomized Design (CRD) is the simplest type of experimental design.
- In this design, the treatments are assigned completely at random to all the experimental units, so that every unit has an equal chance of receiving any treatment.
- It assumes that all experimental units are homogeneous — meaning they are similar in all aspects except for the treatment applied.
- Purpose: To test whether there are significant differences between treatment means under conditions of minimum bias and maximum randomness.

Key Characteristics

Feature	**Description**
Structure	Treatments are randomly assigned to experimental units.
Homogeneity	All units are assumed to be uniform (no blocking needed).
Flexibility	Any number of treatments and replications can be used.
Error Control	Randomization helps control experimental error due to unknown factors.

Completely randomized design

- CRD is the basic single factor design.
- In this design the treatments are assigned completely at random so that each experimental unit has the same chance of receiving any one treatment.
- But CRD is appropriate only when the experimental material is homogeneous.
- As there is generally large variation among experimental plots due to many factors CRD is not preferred in field experiments.
- In laboratory experiments and greenhouse studies it is easy to achieve homogeneity of experimental materials and therefore CRD is most useful in such experiments.

Layout of a CRD

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Layout of a CRD

t = number of treatments

r = number of replications per treatment

$N = t \times r$ = total number of observations

y_{ij} = observation of the ith treatment in the jth replication

$T_i = \sum_{j=1}^{r} y_{ij}$ = total for treatment i

$G = \sum_{i=1}^{t} \sum_{j=1}^{r} y_{ij}$ = grand total

- **Step 1:** Correction Factor (CF): Used to simplify sum of squares calculations. $CF = \frac{G^2}{N}$
- **Step 2:** Total Sum of Squares (SSTotal): Measures overall variation in all data. $SS_{Total} = \sum y_{ij}^2 - CF$
- **Step 3:** Treatment Sum of Squares (SSTreat): Measures variation due to treatments. $SS_{Treat} = \sum_{i=1}^{t} \frac{T_i^2}{r} - CF$
- **Step 4:** Error Sum of Squares (SSE): Measures random (unexplained) variation $SS_{Error} = SS_{Total} - SS_{Treat}$
- **Step 5:** Degrees of Freedom (df)

Source	**Degrees of Freedom**
Treatments	(t - 1)
Error	(N - t)
Total	(N - 1)

Layout of a CRD

$$MS_{Treat} = \frac{SS_{Treat}}{t-1}$$

- **Step 6:** Mean Squares (MS): $MS_{Error} = \frac{SS_{Error}}{N-t}$
- **Step 7:** F-ratio: Tests whether treatments differ significantly. $F = \frac{MS_{Treat}}{MS_{Error}}$

  Compare this F with the tabulated value $F_{\alpha,(t-1),(N-t)}$.

  If $F_{cal} > F_{tab}$: reject H_0 (treatments differ significantly).
- **Step 8:** ANOVA Table Structure

Source	df	SS	MS	F
Treatments	$t-1$	SS_{Treat}	MS_{Treat}	MS_{Treat}/MS_E
Error	$N-t$	SS_E	MS_E	—
Total	$N-1$	SS_T	—	—

- **Step 9:**
  - Null hypothesis (H_0): (No difference among treatment means)
  - Alternative hypothesis (H_1): At least one treatment mean differs.

CRD Problem

1. Three fertilizers A, B, C are tested on a homogeneous field. Each treatment is replicated 4 times (so t=3, r=4, total N=12). Yields (kg) are: A: 18, 20, 19, 22; B: 24, 22, 21, 23; C: 26, 25, 27, 28 , Test at α=0.05, whether there are significant treatment differences.

Step 1 — totals and grand total:

- $T_A = 18 + 20 + 19 + 22 = 79$
- $T_B = 24 + 22 + 21 + 23 = 90$
- $T_C = 26 + 25 + 27 + 28 = 106$
- Grand total $G = 79 + 90 + 106 = 275$
- $N = 12, r = 4$

Step 2 — Total sum of squares (using totals-of-squares formula)

$$SS_{Total} = \sum y_{ij}^2 - \frac{G^2}{N}$$

Compute $\sum y_{ij}^2$: $18^2 + 20^2 + \cdots + 28^2 =$ (sum of squares of all 12 observations) = **(calculated)** (we use the final numeric)

$$SS_{Total} = \sum y^2 - \frac{275^2}{12} = 110.9166667$$

Step 3 — Treatment sum of squares

$$SS_{Treat} = \sum \frac{T_i^2}{r} - \frac{G^2}{N} = \frac{79^2 + 90^2 + 106^2}{4} - \frac{275^2}{12} = 92.1666667$$

Step 4 — Error sum of squares

$$SS_E = SS_{Total} - SS_{Treat} = 110.9166667 - 92.1666667 = 18.75$$

CRD Problem

Step 5 — Degrees of freedom and mean squares

- $df_{Treat} = t - 1 = 2$
- $df_E = N - t = 12 - 3 = 9$
- $MS_{Treat} = SS_{Treat}/2 = 92.1666667/2 = 46.0833333$
- $MS_E = SS_E/9 = 18.75/9 = 2.0833333$

Step 6 — F statistic

$$F = \frac{MS_{Treat}}{MS_E} = \frac{46.0833333}{2.0833333} \approx 22.12$$

Step 7 — Decision at $\alpha = 0.05$

- df numerator = 2, df denominator = 9. Critical $F_{0.05,2,9} \approx 4.26$.
- Observed $F \approx 22.12 > 4.26$ → **reject** H_0. There is a significant difference among fertilizers.

ANOVA summary (rounded):

Source	df	SS	MS	F
Treatments	2	92.1667	46.0833	22.12
Error	9	18.7500	2.0833	
Total	11	110.9167		

Conclusion: Fertilizers differ significantly in mean yield.

Critical values of F for the 0.05 significance level:

	1	2	3	4	5	6
1	161.45	199.50	215.71	224.58	230.16	233.99
2	18.51	19.00	19.16	19.25	19.30	19.33
3	10.13	9.55	9.28	9.12	9.01	8.94
4	7.71	6.94	6.59	6.39	6.26	6.16
5	6.61	5.79	5.41	5.19	5.05	4.95
6	5.99	5.14	4.76	4.53	4.39	4.28
7	5.59	4.74	4.35	4.12	3.97	3.87
8	5.32	4.46	4.07	3.84	3.69	3.58
9	5.12	4.26	3.86	3.63	3.48	3.37
10	4.97	4.10	3.71	3.48	3.33	3.22
11	4.84	3.98	3.59	3.36	3.20	3.10
12	4.75	3.89	3.49	3.26	3.11	3.00
13	4.67	3.81	3.41	3.18	3.03	2.92
14	4.60	3.74	3.34	3.11	2.96	2.85
15	4.54	3.68	3.29	3.06	2.90	2.79
16	4.49	3.63	3.24	3.01	2.85	2.74
17	4.45	3.59	3.20	2.97	2.81	2.70
18	4.41	3.56	3.16	2.93	2.77	2.66
19	4.38	3.52	3.13	2.90	2.74	2.63
20	4.35	3.49	3.10	2.87	2.71	2.60
21	4.33	3.47	3.07	2.84	2.69	2.57
22	4.30	3.44	3.05	2.82	2.66	2.55
23	4.28	3.42	3.03	2.80	2.64	2.53
24	4.26	3.40	3.01	2.78	2.62	2.51
25	4.24	3.39	2.99	2.76	2.60	2.49
26	4.23	3.37	2.98	2.74	2.59	2.47
27	4.21	3.35	2.96	2.73	2.57	2.46
28	4.20	3.34	2.95	2.71	2.56	2.45
29	4.18	3.33	2.93	2.70	2.55	2.43
30	4.17	3.32	2.92	2.69	2.53	2.42
31	4.16	3.31	2.91	2.68	2.52	2.41
32	4.15	3.30	2.90	2.67	2.51	2.40
33	4.14	3.29	2.89	2.66	2.50	2.39

Randomized Block Design (RBD)

- The Randomized Block Design (RBD) is an improvement over the Completely Randomized Design (CRD).
- In RBD, the experimental units are divided into homogeneous groups called blocks, and the treatments are randomly assigned within each block.
- This design helps control the effect of nuisance factors (sources of variation other than the treatments) that can influence the response.
- Variations between blocks are known and controllable.
- Randomization occurs within each block.
- Each treatment appears once in every block.
- Each block contains all treatments, but their order within each block is randomized.
- Purpose: To reduce experimental error by accounting for known sources of variation among experimental units.

Layout of RBD

- t: number of treatments
- r: number of blocks (replications)
- $N = t \times r$: total number of observations
- y_{ij}: observation of ith treatment in j^{th} block
- $T_i = \sum_{j=1}^{r} y_{ij}$: total for treatment i
- $B_j = \sum_{i=1}^{t} y_{ij}$: total for block j
- $G = \sum_{i=1}^{t} \sum_{j=1}^{r} y_{ij}$: grand total

➢ **Step 1:** Correction Factor (CF): Used to simplify sum of squares calculations. $CF = \frac{G^2}{N}$

➢ **Step 2:** Total Sum of Squares (SSTotal): Measures overall variation in all data $SS_{Total} = \sum y_{ij}^2 - CF$

➢ **Step 3:** Treatment Sum of Squares (SSTreat): Measures variation due to treatments. $SS_{Treat} = \sum_{i=1}^{\iota} \frac{T_i^2}{r} - CF$

➢ **Step 4:** Error Sum of Squares (SSE): Measures random (unexplained) variation $SS_{Error} = SS_{Total} - SS_{Treat}$

➢ **Step 5:** Block Sum of Squares (SSBlock): Measures variation due to block effects. $SS_{Block} = \frac{\sum B_j^2}{t} - CF$

➢ **Step 6:** Error Sum of Squares (SSE): Measures unexplained random variation. $SS_E = SS_{Total} - SS_{Treat} - SS_{Block}$

- **Step 7:** Degrees of Freedom (df)

Source	df
Treatments	(t - 1)
Blocks	(r - 1)
Error	((t - 1)(r - 1))
Total	(N - 1)

- **Step 8:** Mean Squares (MS

$$MS_{Treat} = \frac{SS_{Treat}}{t-1}$$

$$MS_{Block} = \frac{SS_{Block}}{r-1}$$

$$MS_E = \frac{SS_E}{(t-1)(r-1)}$$

- **Step 9:** F-ratio: Tests whether treatments differ significantly.

$$F = \frac{MS_{Treat}}{MS_{Error}}$$

Compare F_{cal} with $F_{tab(\alpha,t-1,(t-1)(r-1))}$.

If $F_{cal} > F_{tab}$, reject H_0: treatments differ significantly

- **Step 10:** ANOVA Table Structure

Source	df	SS	MS	F
Treatments	$t-1$	SS_{Treat}	MS_{Treat}	MS_{Treat}/MS_E
Blocks	$r-1$	SS_{Block}	MS_{Block}	—
Error	$(t-1)(r-1)$	SS_E	MS_E	—
Total	$N-1$	SS_{Total}	—	—

- **Step 11:**
  - Null hypothesis (H_0): (No difference among treatment means)
  - Alternative hypothesis (H_1): At least one treatment mean differs.

RBD Problem

2. Three fertilizers A, B, C are tested in 4 blocks to control a one-directional field gradient. Each block contains all three treatments (so t=3, r=4, N=12). Observed yields:

Block	A	B	C
1	18	22	25
2	20	23	26
3	19	24	27
4	21	25	28

Test for treatment effects at α=0.05

Step 1 — Grand total:

- Sum of all observations $G = 278$.

Step 2 — Treatment totals (summing down columns):

- $T_A = 18 + 20 + 19 + 21 = 78$
- $T_B = 22 + 23 + 24 + 25 = 94$
- $T_C = 25 + 26 + 27 + 28 = 106$

Step 3 — Block totals (summing across rows):

- $B_1 = 18 + 22 + 25 = 65$
- $B_2 = 20 + 23 + 26 = 69$
- $B_3 = 19 + 24 + 27 = 70$
- $B_4 = 21 + 25 + 28 = 74$

Step 4 — Sum of squares:

$$SS_{Total} = \sum y_{ij}^2 - \frac{G^2}{N} = 113.6666667$$

$$SS_{Treat} = \sum \frac{T_i^2}{r} - \frac{G^2}{N} = \frac{78^2 + 94^2 + 106^2}{4} - \frac{278^2}{12} = 98.6666667$$

$$SS_{Block} = \sum \frac{B_j^2}{t} - \frac{G^2}{N} = \frac{65^2 + 69^2 + 70^2 + 74^2}{\downarrow 3} - \frac{278^2}{12} = 13.6666667$$

$$SS_E = SS_{Total} - SS_{Treat} - SS_{Block} = 113.6666667 - 98.6666667 - 13.6666667 = 1.3333333$$

RBD Problem

Step 5 — Degrees and means:

- $df_{Treat} = t - 1 = 2$
- $df_{Block} = r - 1 = 3$
- $df_E = (t-1)(r-1) = 2 \times 3 = 6$
- $MS_{Treat} = 98.6667/2 = 49.3333$
- $MS_E = 1.3333333/6 = 0.2222222$

Step 6 — F statistic

$$F = \frac{49.3333}{0.2222222} \approx 222.00$$

Step 7 — Decision

- df num = 2, df den = 6. Critical $F_{0.05,2,6} \approx 5.14$.
- Observed $F \approx 222 \gg 5.14 \rightarrow$ **reject** H_0. Strong evidence of treatment differences.

ANOVA summary (rounded):

Source	df	SS	MS	F
Treatments	2	98.6667	49.3333	222.00
Blocks	3	13.6667	4.5556	
Error	6	1.3333	0.2222	
Total	11	113.6667		

Critical values of F for the 0.05 significance

	1	2	3	4	5	23
1	161.45	199.50	215.71	224.58	230.16	
2	18.51	19.00	19.16	19.25	19.30	1
3	10.13	9.55	9.28	9.12	9.01	
4	7.71	6.94	6.59	6.39	6.26	
5	6.61	5.79	5.41	5.19	5.05	
6	5.99	5.14	4.76	4.53	4.39	
7	5.59	4.74	4.35	4.12	3.97	
8	5.32	4.46	4.07	3.84	3.69	
9	5.12	4.26	3.86	3.63	3.48	
10	4.97	4.10	3.71	3.48	3.33	
11	4.84	3.98	3.59	3.36	3.20	
12	4.75	3.89	3.49	3.26	3.11	
13	4.67	3.81	3.41	3.18	3.03	
14	4.60	3.74	3.34	3.11	2.96	
15	4.54	3.68	3.29	3.06	2.90	
16	4.49	3.63	3.24	3.01	2.85	
17	4.45	3.59	3.20	2.97	2.81	
18	4.41	3.56	3.16	2.93	2.77	
19	4.38	3.52	3.13	2.90	2.74	
20	4.35	3.49	3.10	2.87	2.71	
21	4.33	3.47	3.07	2.84	2.69	
22	4.30	3.44	3.05	2.82	2.66	
23	4.28	3.42	3.03	2.80	2.64	
24	4.26	3.40	3.01	2.78	2.62	
25	4.24	3.39	2.99	2.76	2.60	
26	4.23	3.37	2.98	2.74	2.59	
27	4.21	3.35	2.96	2.73	2.57	
28	4.20	3.34	2.95	2.71	2.56	
29	4.18	3.33	2.93	2.70	2.55	
30	4.17	3.32	2.92	2.69	2.53	
31	4.16	3.31	2.91	2.68	2.52	
32	4.15	3.30	2.90	2.67	2.51	
33	4.14	3.29	2.89	2.66	2.50	

Conclusion: Fertilizers differ significantly in mean yield.

RBD Problem

2. The following data is related to the measurement of cholesterol content (%) performed in 4 different laboratories w.r.t 4 diet foods. The design adopted is a Randomized Block design. Analyze the data and comment on your result [Given F(0.05) =3.86 for (3,6) D.F]:

Laboratory	D1	D2	D3	D4
A	2	5	4	3
B	3	7	4	2
C	5	5	6	4
D	5	5	8	5

Step 1: Identify design parameters

- Treatments (diets) $t = 4$
- Blocks (laboratories) $r = 4$
- Total observations $N = t \cdot r = 16$

Step 2: Compute totals

Treatment totals (sum of columns)

$$T_1 = 2 + 3 + 5 + 5 = 15$$
$$T_2 = 5 + 7 + 5 + 5 = 22$$
$$T_3 = 4 + 4 + 6 + 8 = 22$$
$$T_4 = 3 + 2 + 4 + 5 = 14$$

$$G = \text{Grand total} = 15 + 22 + 22 + 14 = 73$$

Block totals (sum of rows)

$$B_A = 2 + 5 + 4 + 3 = 14$$
$$B_B = 3 + 7 + 4 + 2 = 16$$
$$B_C = 5 + 5 + 6 + 4 = 20$$
$$B_D = 5 + 5 + 8 + 5 = 23$$

Step 3: Compute sum of squares

Total sum of squares

$$SS_{Total} = \sum y_{ij}^2 - \frac{G^2}{N}$$

Compute $\sum y_{ij}^2$:

$$2^2 + 5^2 + 4^2 + 3^2 + 3^2 + 7^2 + 4^2 + 2^2 + 5^2 + 5^2 + 6^2 + 4^2 + 5^2 + 5^2 + 8^2 + 5^2$$

Stepwise:

- Row A: $2^2 + 5^2 + 4^2 + 3^2 = 4 + 25 + 16 + 9 = 54$
- Row B: $3^2 + 7^2 + 4^2 + 2^2 = 9 + 49 + 16 + 4 = 78$
- Row C: $5^2 + 5^2 + 6^2 + 4^2 = 25 + 25 + 36 + 16 = 102$
- Row D: $5^2 + 5^2 + 8^2 + 5^2 = 25 + 25 + 64 + 25 = 139$

$\sum y_{ij}^2 = 54 + 78 + 102 + 139 = 373$

$$SS_{Total} = 373 - \frac{73^2}{16} = 373 - \frac{5329}{16} = 373 - 333.0625 = 39.9375$$

Treatment sum of squares

$$SS_{Treat} = \frac{\sum T_i^2}{r} - \frac{G^2}{N}$$

$$\sum T_i^2 = 15^2 + 22^2 + 22^2 + 14^2 = 225 + 484 + 484 + 196 = 1389$$

$$SS_{Treat} = \frac{1389}{4} - \frac{5329}{16} = 347.25 - 333.0625 = 14.1875$$

Block sum of squares

$$SS_{Block} = \frac{\sum B_j^2}{t} - \frac{G^2}{N}$$

$$\sum B_j^2 = 14^2 + 16^2 + 20^2 + 23^2 = 196 + 256 + 400 + 529 = 1381$$

$$SS_{Block} = \frac{1381}{4} - 333.0625 = 345.25 - 333.0625 = 12.1875$$

Error sum of squares

$$SS_E = SS_{Total} - SS_{Treat} - SS_{Block} = 39.9375 - 14.1875 - 12.1875 = 13.5625$$

Step 4: Degrees of freedom

Source	df
Treatments	t-1 = 3
Blocks	r-1 = 3
Error	(t-1)(r-1) = 9
Total	N-1 = 15

Step 5: Mean squares

$$MS_{Treat} = \frac{SS_{Treat}}{df_{Treat}} = \frac{14.1875}{3} = 4.7292$$

$$MS_E = \frac{SS_E}{df_E} = \frac{13.5625}{9} = 1.5069$$

$$MS_{Block} = \frac{SS_{Block}}{df_{Block}} = \frac{12.1875}{3} = 4.0625$$

Step 6: F-statistic

$$F_{obs} = \frac{MS_{Treat}}{MS_E} = \frac{4.7292}{1.5069} \approx 3.14$$

Step 7: Decision

- Given: $F_{0.05,3,6} = 3.86$
- Observed $F = 3.14 < 3.86$ → **Fail to reject H0**

Conclusion: At 5% significance level, there is **no significant difference in cholesterol content among the four diets**, after accounting for laboratory (block) variation.

Step 8: ANOVA Table

Source	df	SS	MS	F
Treatments	3	14.1875	4.7292	3.14
Blocks	3	12.1875	4.0625	—
Error	9	13.5625	1.5069	—
Total	15	39.9375	—	—

Latin Square Design (LSD)

- LSD is an experimental design used when two sources of variation (besides the treatment) affect the response variable.
- It extends the Randomized Block Design by blocking in two directions (rows and columns), controlling variability due to both.
- It is very useful when the experimenter wants to study treatment effects while removing the influence of two nuisance factors.
- **Purpose:** To study the effect of treatments while controlling the variability arising from two different nuisance factors

Key Features

Feature	Description
Blocking	Two blocking factors (rows and columns).
Randomization	Each treatment appears once in each row and once in each column, but order is randomized.
Homogeneity	Each cell represents one experimental unit; all are assumed similar apart from known variations.
Number of Treatments (t)	Equal to the number of rows and columns.

Layout of LSD

p: order of the square (number of rows = columns = treatments)

y_{ijk}: observation for treatment i, in row j, column k

$T_i = \sum y_{ijk}$: total for treatment i

$R_j = \sum y_{ijk}$: total for row j

$C_k = \sum y_{ijk}$: total for column k

$G = \sum y_{ijk}$: grand total

$N = p^2$: total number of observations

- **Step 1:** Correction Factor (CF): Used to simplify sum of squares calculations. $CF = \frac{G^2}{p^2}$
- **Step 2:** Total Sum of Squares (SSTotal): Measures overall variation in all data. $SS_{Total} = \sum y_{ij}^2 - CF$
- **Step 3:** Treatment Sum of Squares (SSTreat): Measures variation due to treatments. $SS_{Treat} = \frac{\sum T_i^2}{p} - CF$
- **Step 4:** Row Sum of Squares (SSRow): $SS_{Row} = \frac{\sum R_j^2}{p} - CF$
- **Step 5:** Column Sum of Squares (SSCol): $SS_{Col} = \frac{\sum C_k^2}{p} - CF$
- **Step 6:** Error Sum of Squares (SSE): Measures unexplained random variation.

$$SS_E = SS_{Total} - SS_{Treat} - SS_{Row} - SS_{Col}$$

- **Step 7:** Degrees of Freedom (df)

Source	df
Treatments	(p - 1)
Rows	(p - 1)
Columns	(p - 1)
Error	((p - 1)(p - 2))
Total	(p^2 - 1)

- **Step 8:** Mean Squares (MS): $MS_{Treat} = \frac{SS_{Treat}}{p-1}$

$$MS_E = \frac{SS_E}{(p-1)(p-2)}$$

- **Step 9:** F-ratio: Tests whether treatments differ significantly. $F = \frac{MS_{Treat}}{MS_{Error}}$

Compare F_{cal} with $F_{tab(\alpha,p-1,(p-1)(p-2))}$

- **Step 10:** ANOVA Table Structure

Source	df	SS	MS	F
Rows	$p-1$	SS_{Row}	MS_{Row}	—
Columns	$p-1$	SS_{Col}	MS_{Col}	—
Treatments	$p-1$	SS_{Treat}	MS_{Treat}	MS_{Treat}/MS_E
Error	$(p-1)(p-2)$	SS_E	MS_E	—
Total	p^2-1	SS_{Total}	—	—

- **Step 11:**
  - Null hypothesis (H_o): (No difference among treatment means)
  - Alternative hypothesis (H_1): At least one treatment mean differs.

LSD Problem

4. Four treatments A, B, C, D are allocated in a 4×4 Latin square to control two nuisance sources (rows and columns). Observations (rows = 1..4, columns = 1..4) with the Latin arrangement :Row 1: A(18), B(22), C(25), D(28), Row 2: B(21), C(24), D(26), A(20), Row 3: C(19), D(23), A(22), B(25), Row 4: D(20), A(21), B(23), C(27). Test at α=0.05 whether treatment differences are significant.

Step 1 — Grand total:

- $G = 364$

Step 2 — Treatment totals (each treatment appears once in each row and column)

- $T_A = 18 + 20 + 22 + 21 = 81$
- $T_B = 22 + 21 + 25 + 23 = 91$
- $T_C = 25 + 24 + 19 + 27 = 95$
- $T_D = 28 + 26 + 23 + 20 = 97$

Step 3 — Row totals:

- $R_1 = 18 + 22 + 25 + 28 = 93$
- $R_2 = 21 + 24 + 26 + 20 = 91$
- $R_3 = 19 + 23 + 22 + 25 = 89$
- $R_4 = 20 + 21 + 23 + 27 = 91$

Step 4 — Column totals:

- $C_1 = 18 + 21 + 19 + 20 = 78$
- $C_2 = 22 + 24 + 23 + 21 = 90$
- $C_3 = 25 + 26 + 22 + 23 = 96$
- $C_4 = 28 + 20 + 25 + 27 = 100$

LSD Problem

Step 5 — Sum of squares (using usual formulas for Latin square)

$$SS_{Total} = \sum y^2 - \frac{G^2}{p^2} = 127.0$$

$$SS_{Treat} = \sum \frac{T_i^2}{p} - \frac{G^2}{p^2} = \frac{81^2 + 91^2 + 95^2 + 97^2}{4} - \frac{364^2}{16} = 38.0$$

$$SS_{Row} = \sum \frac{R_j^2}{p} - \frac{G^2}{p^2} = \frac{93^2 + 91^2 + 89^2 + 91^2}{4} - \frac{364^2}{16} = 2.0$$

$$SS_{Col} = \sum \frac{C_j^2}{p} - \frac{G^2}{p^2} = \frac{78^2 + 90^2 + 96^2 + 100^2}{4} - \frac{364^2}{16} = 69.0$$

$$SS_E = SS_{Total} - SS_{Treat} - SS_{Row} - SS_{Col} = 127.0 - 38.0 - 2.0 - 69.0 = 18.0$$

Step 6 — Degrees and means:

- $p = 4$. df$_{Rows}$=3, df$_{Cols}$=3, df$_{Treat}$=3, df$_E$ = $(p-1)(p-2) = 3 \times 2 = 6$.
- $MS_{Treat} = 38.0/3 = 12.6667$
- $MS_E = 18.0/6 = 3.0$

Step 7 — F statistic

$$F = \frac{MS_{Treat}}{MS_E} = \frac{12.6667}{3.0} \approx 4.222$$

Step 8 — Decision

- df num = 3, df den = 6. Critical $F_{0.05,3,6} \approx 4.76$.
- Observed $F \approx 4.222 < 4.76$ → **fail to reject** H_0 at 5% level. No statistically significant treatment differences at $\alpha = 0.05$.

ANOVA summary (rounded):

Source	df	SS	MS	F
Rows	3	2.0	0.6667	
Columns	3	69.0	23.0000	
Treatments	3	38.0	12.6667	4.222
Error	6	18.0	3.0000	
Total	15	127.0		

Conclusion: After adjusting for row and column effects, there is **no significant treatment effect at 5%** (observed F slightly below critical).

Critical values of F for the 0.05

	1	2	3	4
1	161.45	199.50	215.71	224.58
2	18.51	19.00	19.16	19.25
3	10.13	9.55	9.28	9.12
4	7.71	6.94	6.59	6.39
5	6.61	5.79	5.41	5.19
6	5.99	5.14	4.76	4.53
7	5.59	4.74	4.35	4.12
8	5.32	4.46	4.07	3.84
9	5.12	4.26	3.86	3.63
10	4.97	4.10	3.71	3.48
11	4.84	3.98	3.59	3.36
12	4.75	3.89	3.49	3.26
13	4.67	3.81	3.41	3.18
14	4.60	3.74	3.34	3.11
15	4.54	3.68	3.29	3.06
16	4.49	3.63	3.24	3.01
17	4.45	3.59	3.20	2.97
18	4.41	3.56	3.16	2.93
19	4.38	3.52	3.13	2.90
20	4.35	3.49	3.10	2.87
21	4.33	3.47	3.07	2.84
22	4.30	3.44	3.05	2.82
23	4.28	3.42	3.03	2.80
24	4.26	3.40	3.01	2.78
25	4.24	3.39	2.99	2.76
26	4.23	3.37	2.98	2.74
27	4.21	3.35	2.96	2.73
28	4.20	3.34	2.95	2.71
29	4.18	3.33	2.93	2.70
30	4.17	3.32	2.92	2.69
31	4.16	3.31	2.91	2.68
32	4.15	3.30	2.90	2.67
33	4.14	3.29	2.89	2.66

Full factorial design

- A Full Factorial Design is an experimental setup where all possible combinations of factors and their levels are tested.
- It allows the experimenter to study: the individual effects (main effects) of each factor, interaction effects between factors.
- It is one of the most informative and complete designs used in industrial, agricultural, and scientific research.
- Purpose: To evaluate how two or more factors simultaneously affect a response variable and to understand interactions among them.

2^2 **design** → 2 factors, 2 levels → 4 runs

2^3 **design** → 3 factors, 2 levels → 8 runs

3^2 **design** → 2 factors, 3 levels → 9 runs

Key Features

Feature	**Description**
Factors	Independent variables (e.g., temperature, pressure, time).
Levels	Different settings of each factor (e.g., high/low, 2 or 3 levels).
Design Type	Every possible combination of factor levels is tested.
Information Obtained	Main effects and interaction effects.

Full factorial design -2^2, 2^3, 2^k

2^2 Design (2 factors, 2 levels)

Factors: A and B

Levels: Low (-1), High (+1)

Number of experiments: $2^2 = 4$

Run	A	B
1	-1	-1
2	+1	-1
3	-1	+1
4	+1	+1

- Purpose: Study the main effects of A and B, and their interaction effect (A×B)
- Example: Effect of Temperature (A) and Pressure (B) on yield.

2^3 Design (3 factors, 2 levels)

Factors: A, B, C

Levels: Low (-1), High (+1)

Number of experiments: $2^3 = 8$

Run	A	B	C
1	-	-	-
2	+	-	-
3	-	+	-
4	+	+	-
5	-	-	+
6	+	-	+
7	-	+	+
8	+	+	+

- Purpose: Study main effects (A, B, C), two-factor interactions (AB, AC, BC), and three-factor interaction (ABC).
- Example: Studying Temperature (A), Pressure (B), and Catalyst Amount (C) on reaction yield.

Full factorial design - 2^2, 2^3, 2^k

2^k Design (k factors, 2 levels each)

Factors: $A_1, A_2, \ldots, A_k$

Levels: Low (-1), High (+1)

Number of experiments: 2^k

Main effects: Each factor individually

Interaction effects: All possible combinations of factors (two-factor, three-factor, ... k-factor)

- As k increases, number of runs doubles for each extra factor.
- Example: 5 factors in a chemical process → 32 experiments to test all main and interaction effects.

Design	Factors	Levels	Runs	Effects Studied
2^2	2	2	4	2 main, 1 interaction
2^3	3	2	8	3 main, 3 two-factor, 1 three-factor
2^k	k	2	2^k	k main + all possible interactions

Problem in Full factorial design

5. A chemical engineer studies the effect of Temperature (A) and Pressure (B) on product yield. Both factors are at two coded levels: low = −1, high = +1. Each treatment combination is run twice (two replicates). The observed yields (%) are:

A	B	Run 1	Run 2
−1	−1	42	43
+1	−1	50	49
−1	+1	46	45
+1	+1	62	61

- **Perform a full 2^2 factorial analysis: estimate main effects and interaction, compute sums of squares, build the ANOVA table, and test treatment effects at α=0.05.**

1. Notation & basics

- $k = 2$ factors → $2^k = 4$ treatment combinations.
- Replicates $r = 2$. Total runs $N = 2^k \times r = 4 \times 2 = 8$.
- Label combinations and list all 8 observations (in run order):

1. $A = -1, B = -1$: 42
2. $A = -1, B = -1$: 43
3. $A = +1, B = -1$: 50
4. $A = +1, B = -1$: 49
5. $A = -1, B = +1$: 46
6. $A = -1, B = +1$: 45
7. $A = +1, B = +1$: 62
8. $A = +1, B = +1$: 61

Grand total:

$$G = 42 + 43 + 50 + 49 + 46 + 45 + 62 + 61 = 398$$

Number of observations:

$$N = 8$$

2. Compute simple effect contrasts (using ±1 coding)

Define the contrast for a factor as sum of $x_i \cdot y_i$, where x_i is the factor code (+1 for high, −1 for low).

For factor A:

- Sum of responses at $A = +1$: runs 3,4,7,8 → $50 + 49 + 62 + 61 = 222$.
- Sum at $A = -1$: runs 1,2,5,6 → $42 + 43 + 46 + 45 = 176$.
- Contrast $C_A =$ (sum at $A+$) − (sum at $A-$) $= 222 - 176 = 46$.

Estimated main effect (standard 2^k formula):

$$\hat{A} = \frac{C_A}{2^{k-1} \cdot r/r_norm} = \frac{C_A}{2^{k-1}} \quad \text{(common practical formula)}$$

For $k = 2$, divisor $2^{k-1} = 2$. But note when replicates exist we usually quote effect = $C_A/(2^{k-1} \cdot r/r)$ — simpler:

$$\hat{A} = \frac{222 - 176}{4} = \frac{46}{4} = 11.5.$$

(Explanation: each effect equals difference of means; dividing by $2^{k-1} = 2$ on averages vs dividing by 4 when using sums — both consistent; here we used sums divided by 4.)

For factor B:

- Sum at $B = +1$: runs 5,6,7,8 → $46 + 45 + 62 + 61 = 214$.
- Sum at $B = -1$: runs 1,2,3,4 → $42 + 43 + 50 + 49 = 184$.
- Contrast $C_B = 214 - 184 = 30$.

$$\hat{B} = \frac{30}{4} = 7.5.$$

For interaction AB:

- AB = +1 when A and B have same sign: combinations (−1,−1) and (+1,+1). Sum for AB=+1: runs 1,2,7,8 → $42 + 43 + 62 + 61 = 208$.
- AB = −1 when signs differ: runs 3,4,5,6 → $50 + 49 + 46 + 45 = 190$.
- Contrast $C_{AB} = 208 - 190 = 18$.

$$\widehat{AB} = \frac{18}{4} = 4.5.$$

Estimated effects (rounded):

- $A = 11.5$
- $B = 7.5$
- $AB = 4.5$

Interpretation: Increasing A (low → high) raises yield by about 11.5% on average; increasing B raises yield by about 7.5%; small positive interaction (4.5).

3. Sum of squares via contrasts (standard 2^k formula with replicates)

For a 2^k design with r replicates, the sum of squares for a contrast C is:

$$SS = \frac{C^2}{2^k\, r} \quad (\text{where } 2^k \text{ is number of treatment combinations}).$$

Here $2^k = 4,\ r = 2$, so denominator = $4 \times 2 = 8$.

Calculate each SS:

- $SS_A = \frac{C_A^2}{8} = \frac{46^2}{8} = \frac{2116}{8} = 264.5.$
- $SS_B = \frac{30^2}{8} = \frac{900}{8} = 112.5.$
- $SS_{AB} = \frac{18^2}{8} = \frac{324}{8} = 40.5.$

Total sum of squares (SSTotal)

$$SS_{Total} = \sum y_i^2 - \frac{G^2}{N}.$$

Compute $\sum y_i^2$ (each squared, then summed):

- $42^2 = 1764$
- $43^2 = 1849$ → partial sum = $1764 + 1849 = 3613$
- $50^2 = 2500$ → partial sum = $3613 + 2500 = 6113$
- $49^2 = 2401$ → $6113 + 2401 = 8514$
- $46^2 = 2116$ → $8514 + 2116 = 10630$
- $45^2 = 2025$ → $10630 + 2025 = 12655$
- $62^2 = 3844$ → $12655 + 3844 = 16499$
- $61^2 = 3721$ → $16499 + 3721 = 20220$

So $\sum y_i^2 = 20220.$

Correction factor:

$$CF = \frac{G^2}{N} = \frac{398^2}{8} = \frac{158404}{8} = 19800.5.$$

Thus

$$SS_{Total} = 20220 - 19800.5 = 419.5.$$

Error (residual) sum of squares

$$SS_E = SS_{Total} - (SS_A + SS_B + SS_{AB}).$$

Compute sum of treatment SS:

$$SS_A + SS_B + SS_{AB} = 264.5 + 112.5 + 40.5 = 417.5.$$

So residual:

$$SS_E = 419.5 - 417.5 = 2.0.$$

(That small SSE arises because replicates were very consistent.)

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4. Degrees of freedom and mean squares

Degrees:

- df_A = 1
- df_B = 1
- df_{AB} = 1
- df_E = $N - 2^k = 8 - 4 = 4$ (or df_{Total} = 7, so 7 − 3 = 4)
- df_{Total} = 8 − 1 = 7

Mean squares:

- $MS_A = SS_A/1 = 264.5.$
- $MS_B = 112.5.$
- $MS_{AB} = 40.5.$
- $MS_E = SS_E/4 = 2.0/4 = 0.5.$

5. ANOVA table

Source	df	SS	MS	F (MS/MS_E)
A (Temperature)	1	264.5	264.5	$264.5/0.5 = 529.0$
B (Pressure)	1	112.5	112.5	$112.5/0.5 = 225.0$
AB (Interaction)	1	40.5	40.5	$40.5/0.5 = 81.0$
Error (Residual)	4	2.0	0.5	—
Total	7	419.5	—	—

6. Hypothesis testing ($\alpha = 0.05$)

For each effect: compare observed F with table $F_{0.05,1,4}$.

Critical value $F_{0.05,1,4} \approx 7.708$. (From F-table for df1=1, df2=4.)

- For A: $F = 529.0 \gg 7.708$ → **reject** H_0 for A. Temperature is highly significant.
- For B: $F = 225.0 \gg 7.708$ → **reject** H_0 for B. Pressure is highly significant.
- For AB: $F = 81.0 \gg 7.708$ → **reject** H_0 for interaction. The interaction is also highly significant.

All three (A, B, and AB) are statistically significant at the 5% level.

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7. Final conclusions

- Estimated main effects: $A = 11.5$, $B = 7.5$. Increasing Temperature and Pressure increases yield.
- Interaction $AB = 4.5$ (positive) — the effect of one factor depends on the level of the other.
- ANOVA shows **A, B, and interaction AB are all highly significant** (all F-values ≫ critical value) — so both main effects and their interaction matter for yield.

Critical values of F for the 0.05 s

	1	2	3	4
1	161.45	199.50	215.71	224.58
2	18.51	19.00	19.16	19.25
3	10.13	9.55	9.28	9.12
4	7.71	6.94	6.59	6.39
5	6.61	5.79	5.41	5.19
6	5.99	5.14	4.76	4.53
7	5.59	4.74	4.35	4.12
8	5.32	4.46	4.07	3.84
9	5.12	4.26	3.86	3.63
10	4.97	4.10	3.71	3.48
11	4.84	3.98	3.59	3.36
12	4.75	3.89	3.49	3.26
13	4.67	3.81	3.41	3.18
14	4.60	3.74	3.34	3.11
15	4.54	3.68	3.29	3.06
16	4.49	3.63	3.24	3.01
17	4.45	3.59	3.20	2.97
18	4.41	3.56	3.16	2.93
19	4.38	3.52	3.13	2.90
20	4.35	3.49	3.10	2.87
21	4.33	3.47	3.07	2.84
22	4.30	3.44	3.05	2.82
23	4.28	3.42	3.03	2.80
24	4.26	3.40	3.01	2.78
25	4.24	3.39	2.99	2.76
26	4.23	3.37	2.98	2.74
27	4.21	3.35	2.96	2.73
28	4.20	3.34	2.95	2.71
29	4.18	3.33	2.93	2.70
30	4.17	3.32	2.92	2.69
31	4.16	3.31	2.91	2.68
32	4.15	3.30	2.90	2.67
33	4.14	3.29	2.89	2.66

Drawback of Full Factorial designs

Drawback	Explanation
Rapid increase in number of experiments	The total number of runs grows exponentially with the number of factors: (N = L^k), where *L* = levels and *k* = factors. → e.g., for 6 factors at 3 levels each, (3^6 = 729) experiments — costly and time-consuming.
High cost and time requirement	Performing every combination is often impractical in industrial settings due to resource and labor constraints.
Difficult to analyze higher-order interactions	As the number of factors increases, interpretation of multiple interaction effects becomes complex and often non-intuitive.
Not robust to noise or uncontrollable factors	Full factorial designs focus on main and interaction effects, not on *robustness* against variation (unlike Taguchi designs).
Wastage of effort when higher-order interactions are negligible	In most real-world processes, 3-way or higher interactions are usually insignificant — so studying them adds little value but increases experiment count.
Resource and equipment limitations	Practical constraints such as machine time, material cost, and setup time make large factorial designs infeasible.
Statistical efficiency decreases	When experimental errors are large, collecting too many runs can introduce more variability without improving inference accuracy.

Fractional factorial designs

- A Fractional Factorial Design is a type of experimental design in which only a fraction (subset) of the total number of runs in a full factorial design is conducted.
- It is used when the number of factors is large and performing all possible combinations (as in a full factorial) would be too time-consuming or expensive.
- The idea is that not all high-order interactions (like three-factor or four-factor interactions) are significant; most real-world systems are mainly influenced by main effects and low-order interactions.
- Therefore, by carefully selecting a fraction of the runs, we can still obtain reliable information about the most important effects.
- For example, in a 2^4 full factorial design, there would be 16 runs. A ½ fraction design (2^{4-1}) uses only 8 runs but still provides information about main and two-factor interactions, assuming higher-order interactions are negligible.
- Fractional factorial designs are often represented as $2k-p$, where k is the number of factors and p is the number of factors used to reduce the runs by half, quarter, etc.

Fractional factorial designs

- This design is widely used in industrial experiments, process optimization, and quality improvement (like in Taguchi methods).
- It saves time, cost, and resources, while still giving sufficient insight into which factors significantly affect the outcome.
- However, one limitation is confounding, where some effects may be mixed (aliased) with others, making it harder to interpret results unless the design is chosen carefully.

Taguchi method for optimization

- The Taguchi Method is a statistical approach for optimizing process parameters and improving the quality of manufactured products.
- Robust Design method, also called the Taguchi Method.
- It was developed by Dr. Genichi Taguchi, and it uses a special design of experiments (DOE) called orthogonal arrays to study the effect of multiple variables efficiently.
- Genichi Taguchi (1 January 1924 – 2 June 2012) was a statistician and engineer.
- He obtained his Ph.D. in Statistics and Mathematics Science at Kyushu University University, Japan in 1962.

Robust Design

- Robust design is an “engineering methodology for improving productivity during research and development so that high-quality products can be produced.
- Robust Design is a technique that reduces variation in a product by reducing the sensitivity of the design of the product to sources of variation rather than by controlling their sources.
- The end result is a robust design, a design that has minimum sensitivity to variations in uncontrollable factors
- A system (product or process) is robust if it performs properly in a wide range of conditions.

Robust Products	Products that are not robust
A pen that writes until the ink is empty	Pen that stops writing after a few months
A car that starts at -20 deg	A car that does not start
A vacuum cleaner that maintains suction levels	A vacuum cleaner that loses suction

Robust Design

- In the design of a new product, any design activity can be called robust, if it leads the product;
  - To have longer life (higher reliability)
  - To be more consistent with use
  - To be more consistent from product to product
  - To perform consistently as temperature and other conditions change
- Products and services should be designed to be inherently defect free and high quality
- Meet customer's expectations also under non-ideal conditions
- Disturbances are events that cause the design performance to deviate from its target value
- Taguchi divided disturbances into **three categories**
  - External disturbances: variation in the environment where the product is used
  - Internal disturbances: wear and tear inside a specific unit
  - Disturbances in the production process: deviation from target values

3 Step Method for Achieving Robust Design (Taguchi)

1. Concept Design / System Design

- The process of examining competing technologies for producing a product – Includes choices of technology and process design.
- A prototype design that can be produced and meets customers needs under ideal conditions without disturbances

2. Parameter Design

- The selection of control factors (parameters) and their optimal levels
- The objective is to make the design Robust
- Control factors are those process variables management can influence
- Ex. The procedures used and the type and amount of training
- Often a complex (non-linear) relationship between the control factors and product / design performance
- The optimal parameter levels can be determined through experimentation.

3 Step Method for Achieving Robust Design (Taguchi)

3. Tolerance Design

- Development of specification limits
- Necessary because there will always be some variation in the production process
- Taguchi fiercely advocates aiming for the target value not just settle for inside the specification limits.
- Occurs after the parameter design
- Often results in increased production costs
- More expensive input material might have to be used to meet specifications.

Key Concepts

Term	Description
Control Factors	Variables that can be controlled (e.g., temperature, pressure).
Noise Factors	Variables that cause variability but cannot be easily controlled (e.g., humidity, operator skill).
Orthogonal Array (OA)	A matrix that systematically varies factors and levels while minimizing experiments.
Signal-to-Noise (S/N) Ratio	A measure to determine robustness; larger S/N = better performance.

Steps in Taguchi Method

- Identify the problem → Define objective and performance characteristic.
- Select factors and levels → Choose parameters to vary.
- Select an orthogonal array (OA) → Based on number of factors and levels (e.g., L4, L8, L9).
- Conduct experiments according to OA.
- Calculate S/N ratio for each experiment.
- Analyze results → Identify optimal factor levels.
- Perform confirmation test → Validate improvement.

Signal-to-Noise (S/N) Ratio Formulas

Type	Objective	Formula
Larger-the-better	Maximize performance	$S/N = -10\log_{10}\left(\frac{1}{n}\sum\frac{1}{y_i^2}\right)$
Smaller-the-better	Minimize defects	$S/N = -10\log_{10}\left(\frac{1}{n}\sum y_i^2\right)$
Nominal-the-best	Target value desired	$S/N = 10\log_{10}\left(\frac{\bar{y}^2}{s^2}\right)$

Background of the Taguchi Method

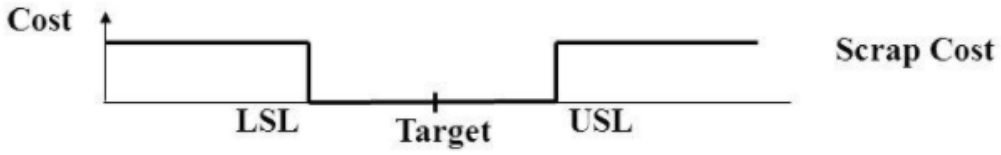
- **Unique aspects of the Taguchi Method**
  - The Taguchi definition of quality
  - The Taguchi Quality Loss Function (QLF)
  - The concept of Robust Design

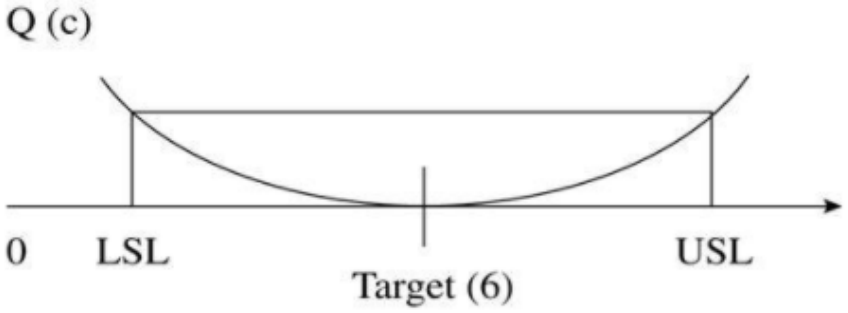
1. The Taguchi definition of quality
   - Ideal quality refers to a target value for determining the quality level
   - Ideal quality is delivered if a product or service tangible performs its intended function throughout its projected life under reasonable operating conditions without harmful side effects
   - Ideal quality is a function of customer perception and satisfaction
   - Service quality is measured in terms of loss to society

Background of the Taguchi Method

2. Taguchi Quality Loss Function (QLF)

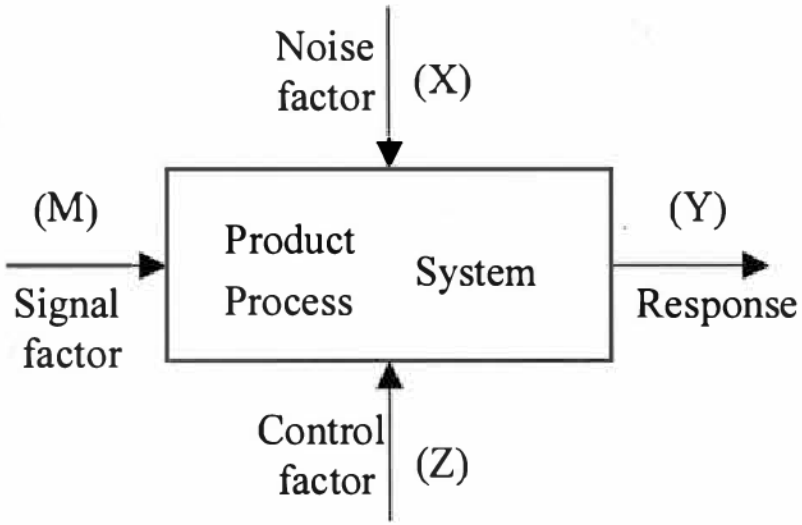
- Taguchi defines Quality as “the loss imparted by the product to society from the time the product is shipped”
- LOSS = Cost to operate, Failure to function, maintenance and repair cost, customer satisfaction, poor design.
- Product to be produced “being within specification





Taguchi Terminology

- A schematic representation of a product / process is shown in a figure. **The response is denoted by y and is called quality characteristics.**



- The parameters that influence the quality characteristic are classified as follows:

- **Signal factor (M):** These are the parameters set by the user or operator of the product **to express the intended value for the response** of the product based on the knowledge of the product being developed.
- Sometimes 2 or more signal factors may also be used in combination to express the desired response.

Taguchi Terminology

- **Noise factor (x):** Certain parameters are not controllable by the designer and are known as noise factors.
- Parameters whose settings (also called levels) are difficult to control in the field or whose levels are expensive to control are also considered noise factors.
- The levels of the noise factors change from one unit to another, from one environment to another, and from time to time.
- **The noise factors cause the response y to deviate from the target specified by the signal factor M and lead to quality loss.**
- **Control factor (z):** These are parameters that can be specified by the designer. **Control factor can take multiple values, called levels.**

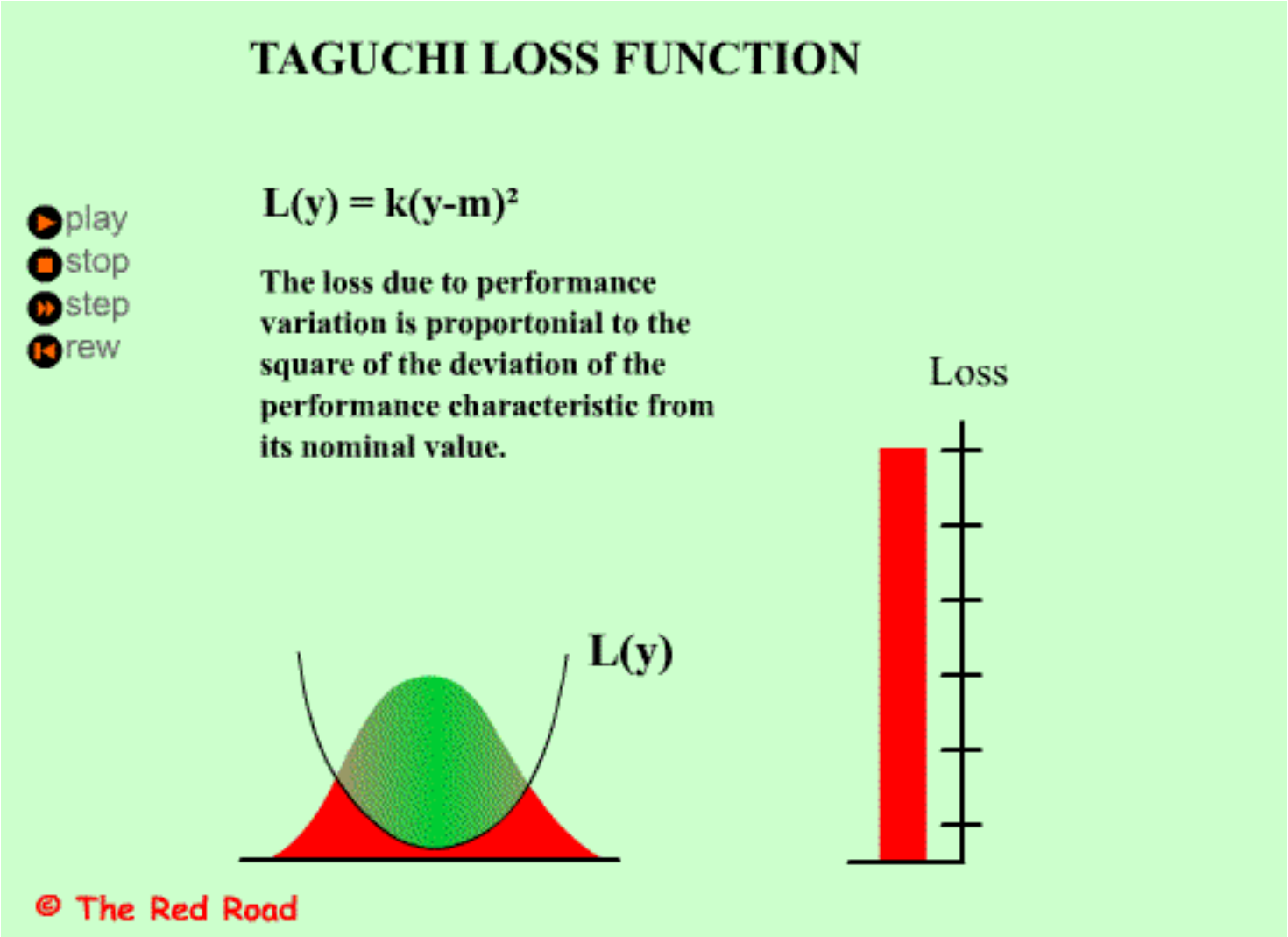
Taguchi's Design of Experiment Idea

- Use **Orthogonal Arrays** that **only estimate main effects and specified interactions.**
- **Intentionally vary the noise factors** so that you choose a set of conditions that will work well in the face of the noise expected in the actual application.
- Inner/Outer Arrays – Two statistical designs per experiment:
  - A design in the control factors called an **inner array**
  - A design in the noise factors called an **outer array**
- Good for estimating C x N Factor Interactions.

Taguchi's Analysis Ideas

- **A single response which makes a tradeoff between setting the mean to a desirable level while keeping the variance low.**
- Always try to **MAXIMIZE** a S/N Ratio
- There are three types:
  - Smaller is Better
  - Target is Best
  - Larger is Better

Taguchi Quality Loss Function (QLF)



Where y is the critical performance parameter value,
L is the loss associated with a particular parameter y,
m is the nominal value of the parameter specification (ie. Target value of y),
k is a constant that depends on the cost at the specification limits. (ie. Quality loss coefficient)

Orthogonal Array

- Instead of testing every possible combination (as in full factorial design), Taguchi introduced Orthogonal Arrays (OAs) — a systematic and balanced way to conduct experiments with minimum number of trials while still capturing the influence of multiple factors.
- An Orthogonal Array (OA) is a matrix that organizes experiments in such a way that the effect of each factor can be independently and efficiently studied.
- Each column of the OA represents a factor, and each row represents a trial (experiment).
- The arrangement is "orthogonal" because the combinations of factor levels are balanced and uncorrelated, allowing fair comparison of factor effects.
- For example, if we have 3 factors, each at 3 levels, a full factorial design will require 3^3 = 27 experiments.
- But using Taguchi's L9 Orthogonal Array, we can study all factors effectively with just 9 experiments.
- Here, each level of each factor appears the same number of times, ensuring balance and orthogonality.

Advantages of Using Orthogonal Arrays

- Reduces the number of experiments drastically
- Efficiently identifies important factors and their optimal levels.
- Robust design — focuses on minimizing variability due to uncontrollable factors ("noise").
- Easy to analyze using Signal-to-Noise (S/N) ratios.

Common Taguchi Orthogonal Arrays

OA Name	No. of Experiments	Max Factors	Levels
L4	4	3	2
L8	8	7	2
L9	9	4	3
L16	16	15	2
L27	27	13	3

Steps in Taguchi Method

1. Choose control factors and their levels
2. Identify uncontrollable (noise) factors and decide on how they will be simulated.
3. Select the response variable(s) and determine the performance measures (mean, standard deviation, SNR, etc.)
4. Setup the experimental layout (choose appropriate design array
5. Conduct the experiments and collect data
6. Analyze the data (effects, ANOVA, regression)
7. Choose optimal control factor levels and predict the performance measure at these levels
8. Confirm the optimal levels by experimentation.

Problems on Taguchi Method

You are studying an alloy's tensile strength (MPa). You want to maximize tensile strength by choosing the best combination of three process factors. Each factor has 3 levels:

Factor A: Temperature — Level 1 = 200°C, Level 2 = 220°C, Level 3 = 240°C

Factor B: Time — Level 1 = 30 min, Level 2 = 45 min, Level 3 = 60 min

Factor C: Catalyst concentration — Level 1 = 1%, Level 2 = 2%, Level 3 = 3%

Use a Taguchi L9 orthogonal array (suitable for 3 factors × 3 levels). Each run is repeated 3 times. Find the factor-level combination that maximizes tensile strength and quantify factor contributions.

L9 layout and measured data

Run	A	B	C	Measurement 1 (MPa)	Measurement 2	Measurement 3
1	1	1	1	480	485	482
2	1	2	2	490	492	491
3	1	3	3	475	478	476
4	2	1	2	500	503	502
5	2	2	3	495	497	496
6	2	3	1	488	486	487
7	3	1	3	505	507	506
8	3	2	1	492	493	491
9	3	3	2	499	501	500

Problems on Taguchi Method

Step 1 — Choose S/N formula

Because we want to **maximize** tensile strength, use the "**larger-the-better**" S/N ratio:

$$\mathrm{S/N}_{LB} = -10\log_{10}\left(\frac{1}{n}\sum_{i=1}^{n}\frac{1}{y_i^2}\right)$$

where y_i are the measured values for a run and n is number of repeats (here $n = 3$).

Step 2 — Compute S/N ratio for each run (example values)

I computed S/N for each run (rounded to 4 decimals):

Run	Mean (MPa)	S/N (dB)
1	482.3333	53.6667
2	491.0000	53.8216
3	476.3333	53.5581
4	501.6667	54.0082
5	496.0000	53.9096
6	487.0000	53.7505
7	506.0000	54.0830
8	492.0000	53.8393
9	500.0000	53.9794

Overall average S/N (grand mean) = **53.8463 dB**.

Step 3 — Factor-level S/N averages (effects)

Average S/N for each level of each factor:

- Factor **A** (Temperature)
  - A Level 1 (200°C): average S/N = **53.6821 dB**
  - A Level 2 (220°C): average S/N = **53.8894 dB**
  - A Level 3 (240°C): average S/N = **53.9672 dB**
- Factor **B** (Time)
  - B Level 1 (30 min): average S/N = **53.9193 dB**
  - B Level 2 (45 min): average S/N = **53.8235 dB**
  - B Level 3 (60 min): average S/N = **53.7053 dB**
- Factor **C** (Catalyst %)
  - C Level 1 (1%): average S/N = **53.7224 dB**
  - C Level 2 (2%): average S/N = **53.9364 dB**
  - C Level 3 (3%): average S/N = **53.8877 dB**

Interpretation: For each factor choose the level with the **highest** average S/N (since larger-the-better).

- Best A level = **3** (240°C)
- Best B level = **1** (30 min)
- Best C level = **2** (2% catalyst)

So Taguchi suggests the optimal combination: **A3, B1, C2** — i.e., **240°C, 30 min, 2% catalyst**.

Step 4 — (Approximate) ANOVA on S/N values — factor contributions

I performed a simple ANOVA using the S/N ratios as response values (this is a common Taguchi approach to estimate relative contributions). The results (rounded):

- Total sum of squares (SS_total) = **0.22907**
- SS(A) = **0.13028** → Contribution ≈ **56.87%**
- SS(B) = **0.03730** → Contribution ≈ **16.28%**
- SS(C) = **0.05098** → Contribution ≈ **22.25%**
- SS(Error) ≈ **0.01052**

Mean squares and F (approx):

- MS_A = 0.06514, MS_B = 0.01865, MS_C = 0.02549
- MS_Error ≈ 0.005258
- F_A ≈ 12.39, F_B ≈ 3.55, F_C ≈ 4.85

Interpretation: Factor **A (Temperature)** is the dominant contributor to variability (~57%), followed by **C (Catalyst)** (~22%), then **B (Time)** (~16%). F-values indicate A is the most statistically influential (for an exact p-value you would compare F to critical F for df; here df are small so treat as illustrative).

Step 5 — Predicted mean performance at the optimum

You can predict the mean tensile strength at the recommended setting by using the main effects on mean response (averages of measured responses). Using the run means:

- Overall mean tensile strength (average of all run-means) = **492.4815 MPa**
- Level mean (A3) = **499.3333 MPa**
- Level mean (B1) = **496.6667 MPa**
- Level mean (C2) = **497.5556 MPa**

A simple additive prediction:

$$\hat{Y}_{A3B1C2} = \bar{Y} + (A3 - \bar{Y}) + (B1 - \bar{Y}) + (C2 - \bar{Y})$$

gives predicted mean tensile strength ≈ **508.59 MPa**.

Run a confirmation experiment at the recommended setting A3-B1-C2 (240°C, 30 min, 2% catalyst). Collect at least 3 repeats and compare the observed mean tensile strength to the predicted ~**508.6 MPa**. If the confirmation run gives comparable improvement, the optimization is validated.